

What Does It Really Cost to Get Sick?

Exercises on the Kibbo Healthcare Cost Dataset (2026): 58 rows, 4 countries (United States, United Kingdom, Germany, Spain), 4 treatments

Teacher edition. This version includes the answer key at the end. Do not hand it to students; a separate student sheet without answers is available.

Level: university (the critical-thinking questions also work for upper high school). **Subjects:** health economics, public policy, statistics, economics. **Tools:** Excel or Google Sheets. No programming needed.

Before you start

The dataset splits cost into layers: **Layer A** is what you pay every year whether you get sick or not (premiums, contributions, taxes); **Layer B** is what a specific treatment costs when you get it; **Layer C** is optional private insurance, shown separately. Read the README first: this dataset deliberately leaves some cells empty, and that is part of what you are going to analyse.

Files used (download links and the README are on the web page of this sheet: getkibbo.com/education/exercises/healthcare-costs):

- [kibbo-healthcare-cost-dataset-2026.csv](#)
- [healthcare-dataset-README.md](#)

Exercises

Exercise 1 — Reading the data (*basic*)

1. Using the “Private insurance claims price” rows, which country is cheapest and which is most expensive for a hip replacement? How many times more expensive is the most expensive than the cheapest?

Exercise 2 — A surprise in the data (*basic*)

1. Look at the private insurance claims price for a knee MRI. Which country is more expensive than the United States, and by what percentage?

Exercise 3 — Billed price vs. cash price (*intermediate*)

1. For the three US treatments that have both a “Listed/billed charge” and a “Cash/self-pay price”, calculate how much lower the cash price is than the billed charge, as a percentage. Which treatment has the biggest gap?

Exercise 4 — What an American worker pays before insurance kicks in (*intermediate*)

- Using the US Layer A rows:
 - What share of the total annual premium does the worker pay?
 - Add the worker's annual premium share and the average annual deductible. What does this simple total represent, and what does it leave out?

Exercise 5 — Comparing the wrong things (*advanced*)

- Spain's total contribution rate is 28.3%. Germany's statutory health insurance contribution is 14.6% plus an average 2.9% additional contribution. A student concludes: "Spanish workers pay almost twice as much for healthcare as German workers." Use the notes column to explain why this conclusion is wrong.

Exercise 6 — Wait times that can't be compared (*advanced*)

- The dataset shows 24.1 days for the US, 42 days for Germany, about 83 days for the UK and 121 days for Spain for an elective hip replacement.
 - Why is it wrong to conclude that the US is fastest and Spain slowest from these four numbers?
 - Which rows can be compared with each other, and what do they show?

Exercise 7 — Missing data as a finding (*advanced*)

- Count the rows where the value is empty. Which countries and which layer are they in? Why does the dataset leave them empty instead of using an estimate, and what does the gap itself tell you about how transparent each health system is?

Exercise 8 — Currency and time (*advanced*)

- The UK hip replacement row shows £6,061 and is labelled "historical reference only". Give two reasons why you shouldn't compare it directly with the US prices in the dataset.

Connect it to real life

What can you do if you think a medical bill is wrong? Use the Kibbo Healthcare Billing Dispute & Refund Letter Generator and the Medical Billing Dispute / Insurance Appeal Kit (getkibbo.com).

Answer key

Exercise 1

- Cheapest: **Spain, \$6,780**. Most expensive: **United States, \$29,006**. $29,006 \div 6,780 = 4.28$ times.

Exercise 2

1. **Germany: \$613 vs. \$566 in the US, +8.3%.** Spain is the cheapest at \$113.

Exercise 3

1. Appendectomy: $(10,972 - 6,848) \div 10,972 = 37.6\%$ lower. Hip replacement: **38.7%** lower. Knee MRI: $(3,653 - 1,595) \div 3,653 = 56.3\%$ lower, the biggest gap.

Exercise 4

- a) $1,440 \div 9,325 = 15.4\%$ (the employer pays the rest).
- b) $\$1,440 + \$1,886 = \$3,326$: roughly what a worker with typical single coverage pays in a year before most insurance cost-sharing applies. It leaves out coinsurance (typically 20% for a hospital admission), copays and anything above the deductible, which can go up to the legal out-of-pocket maximum of **\$9,200 in 2025**.

Exercise 5

1. Spain's 28.3% is **not health-specific**: it also covers sick leave, maternity/paternity and other benefits. Most of it (23.6 points) is paid by the employer; the worker's share is 4.7%. Germany's rate is health insurance only and is split equally between employer and employee. The two numbers measure different things.

Exercise 6

- a) They measure different points in the care pathway. The US and Germany figures are time to a **specialist appointment**; the UK and Spain figures are time to **actual treatment**. The UK figure is also a median across all specialties, not hip-specific.
- b) Comparable time-to-appointment figures: **US 24.1 days, Germany 42 days, Spain 102 days** (Spain's "Time to first specialist consultation" row). The UK doesn't have an equivalent appointment-wait row.

Exercise 7

1. **11 empty values**, all in **Layer B**: 3 for the UK (appendectomy, knee MRI, vaginal delivery), 4 for Germany and 4 for Spain. The public systems' internal per-treatment cost could not be verified from a primary source, and estimating would mean presenting a guess as data. The gap is itself a finding: it is much easier to find out what a US hospital bills for a treatment than what the UK, German or Spanish public systems actually spend on it.

Exercise 8

1. Strong answers mention any two of: it is in **pounds, not dollars**; it is from **2017/18**, so it doesn't reflect inflation since then; it is a **public-system internal cost**, a different kind of figure from a price billed to a patient or insurer.